

Anatomy of Flowering Plants

SHORT NOTES

Anatomy: Anatomy is the study of internal structure of organisms. Plants anatomy include organisation and structure of tissues

Tissues: Tissue is a group of cells having a common origin and usually perform a common function.

Meristematic tissue: The meristematic tissue is made up of the cells which have the capability to divide. Meristems in plants are restricted to a special region and responsible for the growth of plants.

Apical Meristem	Intercalary Meristem	Lateral Meristem
Occurs at the tips of roots and shoots.	Occurs between mature tissue.	Occurs in the mature regions of roots and shoots.
Primary meristem.	Primary meristem.	Secondary meristem.
Increase the length of the plants.	Activity of intercalary meristem adds to the length of plant or its organs.	Appears later than primary meristem and responsible for secondary growth.

Axillary bud: The buds which are present in the axil of leaves and are responsible for forming branches of flowers.

Permanent tissues: The permanent tissues are derived from meristematic tissues and composed of cells, which have lost the ability to divide.

Types of permanent tissues:

- ❖ **Simple:** Parenchyma, collenchyma and sclerenchyma
- ❖ **Complex:** Xylem and phloem
- ❖ **Parenchyma:** Thin walled cells, with intercellular spaces, cell wall is made up of cellulose. It performs the function like photosynthesis, storage, secretion.
- ❖ **Collenchyma:** It is formed of living, closely packed isodimetric cells. It's thickened at the corners due to deposition of cellulose and pectin. It provides mechanical support to the growing parts of the plants.
- ❖ **Sclerenchyma:** It is formed of dead cells with thick and lignified walls. They have two types of cells : Fibres and sclereids.
- ❖ **Xylem:** Xylem consists of tracheids, vessels, xylem fibres and xylem parenchyma. It conducts water and minerals from roots to other parts of the plants.
- ❖ **Protoxylem:** The first formed primary xylem.
- ❖ **Metaxylem:** The later formed primary xylem.
- ❖ **Endarch:** Protoxylem lies towards the centre and metaxylem towards the periphery of the organ.

❖ **Phloem:** Phloem consists of sieve tubes elements, companion cells, phloem fibres and phloem parenchyma. Phloem transports the food material from leaves to various parts of the plant.

❖ **Protophloem:** First formed phloem with the narrow sieve tubes.

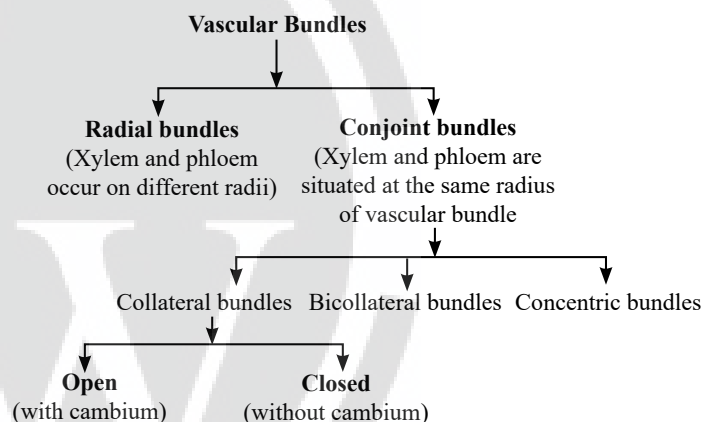
❖ **Metaxylem:** Later formed phloem of with bigger sieve tubes.

The Tissue System:

❖ **Epidermal tissue system:** It includes cuticle, epidermis, epidermal hairs, root hairs, trichomes and stomata.

❖ **The ground tissue system:** It is made up of parenchyma, collenchyma, sclerenchyma. In dicot stems and roots the ground tissue is divided into hypodermis cortex, endodermis, pericycle, medullary rays and pith.

❖ **The vascular tissue system:** It includes vascular bundles which are made up of xylem and phloem.



Dicot Stem	Monocot Stem
The ground tissue is differentiated into cortex, endodermis, pericycle and pith.	The ground tissue is made up of similar cells.
The vascular bundles are arranged in a ring.	The vascular bundles are scattered throughout the ground tissue.
Vascular bundles are open, without bundle sheath and wedge-shaped outline.	Vascular bundles are closed, surrounded by sclerenchymatous bundle sheath, oval or rounded in shape.
The stem shows secondary growth due to presence of cambium between xylem and phloem.	Secondary growth is absent.

Dorsiventral (Dicot) Leaf	Isobilateral (Monocot) Leaf
Stomata are absent or less abundant on the upper side.	The stomata are equally distributed on both sides.
Mesophyll is differentiated into two parts-upper palisade parenchyma and lower spongy parenchyma.	Mesophyll is undifferentiated.
Hypodermis of the midrib region is collenchymatous.	Hypodermis of the midrib region is sclerenchymatous.
Stomata have kidney-shaped guard cells.	Stomata have dumb bell-shaped guard cells.

Dicot root	Monocot root
Cortex is comparatively narrow.	Cortex is very wide.
Endodermis with less thickened casparian stripes are more prominent.	Endodermal cells with highly thickened casparian strips are visible only in young roots.
The xylem and phloem bundles varies from 2 to 5.	Xylem and phloem are more than 6(polyarch condition).
Pith is absent or very small.	Well developed pith is present.
Secondary growth takes place with the help of vascular cambium and cork cambium.	Secondary growth is absent.

